

Business Problem Statement

A large retail company wants a better handle on how its customers shop, so it can grow sales, keep customers satisfied, and build repeat business over time. Management has noticed shifts in buying patterns across age groups, product categories, and channels (online versus in-store), and wants to know what's actually driving those decisions: discounts, reviews, seasonality, or how people prefer to pay.

The task is to analyze the company's consumer behavior dataset to answer this question:

"How can the company use its consumer shopping data to spot trends, improve customer engagement, and sharpen its marketing and product strategies?"

Deliverables

1. **Data preparation and modeling (Python):** Clean and reshape the raw dataset so it's ready for analysis.
2. **Data analysis (SQL):** Structure the data, simulate business transactions, and write queries that show customer segments, loyalty patterns, and what actually drives purchases.
3. **Visualization and insights (Power BI):** Build an interactive dashboard that lets stakeholders see the key patterns and trends for themselves.
4. **Report and presentation:** Write up the key findings and recommendations in a project report, and put together a presentation that walks stakeholders through the insights and what to do about them.
5. **GitHub repository:** Keep all the Python scripts, SQL queries, and dashboard files organized in one repository.